through all events in space that occur at the same time, according to his perspective. As we have done often in previous chapters, we have suppressed one spatial dimension in this diagram for visual clarity. In reality, of course, there is a three-dimensional array of events that occur at the same time according to any observer. Figures 6.8(b) and 6.8(c) give a couple of snapshots at subsequent times—subsequent "slices" of the world-sheet—showing how George sees the two strings approach each other. Of central importance, in Figure 6.8(c) we show the instant in time, according to George, when the two strings first touch and merge together, producing the third string.

Let's now do the same for Gracie. As discussed in Chapter 2, the relative motion of George and Gracie implies that they do not agree on what events occur at the same time. From Gracie's perspective the events in space that occur simultaneously lie on a different plane, as shown in Figure 6.9. That is, from Gracie's perspective, the world-sheet of Figure 6.7(c) must be "sliced" into pieces at a different angle in order to reveal the moment-by-moment progression of the interaction.

In Figures 6.9(b) and 6.9(c) we show subsequent moments in time, now according to Gracie, including the moment when she sees the two incoming strings touch and produce the third string.

By comparing Figures 6.8 (c) and 6.9 (c), as we do in Figure 6.10, we see that George and Gracie do not agree on when and where the two initial strings first touch—where they interact. The string, being an extended object, ensures that there is no unambiguous location in space or moment in time when the strings first interact—rather, it depends upon the state of motion of the observer.

If we apply exactly the same reasoning to the interaction of point particles, as summarized in Figure 6.11, we recover the conclusion stated earlier –there is a definite point in space and moment in time when the point particles interact. Point particles cram all of their interaction into a definite point. When the force involved in an interaction is the gravitational force—that is, when the messenger particle involved in the interaction is the graviton instead of the photon—this complete packing of the force's punch into a single point leads to disastrous results, such as the infinite answers we alluded to earlier. Strings, by contrast, "smear" out the place where interactions occur.